

Negar Honarvar Sedighian

AI Engineering · Computer Vision · Graph Learning

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WORK EXPERIENCE

AI Engineer · Salamat Binesh Farda

Sep 2026–present

Working closely with dentists to develop AI models for dental assistants, integrating clinical expertise into the model development process.

AI Engineer · Amoot

Jan–Sep 2026 · Mashhad

Developed phishing detection systems and contributed to a comprehensive AI assistant fine-tuned for Farsi.

Network Security Assistant · Dotin

Jun–Oct 2023 · Tehran

Managed access to company resources using Windows Azure; identified and analyzed potential network weaknesses and breaches.

Network Security Intern · Dotin School

Jan–Jun 2023 · Tehran

Completed a network security bootcamp covering security vulnerabilities, risk identification, incident response, and penetration testing.

EDUCATION

B.Sc. in Computer Engineering

Sep 2020–Feb 2025

Shahid Beheshti University · Tehran, Iran

Cumulative GPA: **17.49/20 (3.68/4)** · Final two years: **18.32/20 (3.79/4)**

Selected courses (GPA 4/4): Computer Vision, Robotics, Deep Reinforcement Learning, Machine Learning, Artificial Intelligence, Algorithms Design, Computer Simulation, Statistics and Probability, and Data Structures.

High School Diploma in Mathematics

Sep 2017–Jun 2020

Farzanegan 1 High School (SAMPAD) · Mashhad, Iran

GPA: **19.95/20 (4/4)**. Ranked first in the SAMPAD high-school entrance examination in Khorasan Razavi Province (2017).

LANGUAGES

- **English:** Advanced proficiency (CEFR C1); TOEFL iBT **108/120**.
- **French:** Intermediate proficiency (CEFR B1), with ongoing study to strengthen communication skills.
- **Spanish:** Currently developing language proficiency.

RESEARCH EXPERIENCE

B.Sc. Thesis · Cardiac MRI Reconstruction

Image Processing and Distributed Systems Lab · Shahid Beheshti University · Completed

Proposed a cascading dynamic attentive graph neural network for cardiac MRI reconstruction, supervised by Dr. Mohsen Ebrahimi Moghaddam.

SELECTED PROJECTS

DAGMRNet: Dynamic Attentive Graph Learning for Cardiac MRI

Feb 2025

github.com/negarhonorvar/DAGMRNet

- Proposed a dynamic attentive GNN for reconstruction of highly undersampled cardiac MRI images from the CMRxRecon dataset.
- Models spatial and temporal relationships using image self-similarity; uses PromptUnet, introduced in PromptMR, for sensitivity map estimation.

Automated Stock Trading Strategy with Deep Reinforcement Learning

Jun 2024

github.com/negarhonorvar/StockTrading_DRL

- Designed a cascading LSTM-PPO model to capture temporal dependencies in stock data and optimize trading decisions.
- Used a yfinance-based environment with trading data from January 2009 to June 2024.

Enhanced Farsi News Classification

Mar 2024

[GitHub repository](#)

- Developed an enhanced neural network to classify Farsi news articles into categories.
- Prepared the dataset with tokenization and feature extraction.

Quadcopter Guidance for Object Detection

Mar 2024

[GitHub repository](#)

- Designed a controller for autonomous flight over boxes in an urban environment, image capture, and deep-learning-based object recognition.
- Programmed the quadcopter to land beside a matching target and activate its front LEDs.

Bug Algorithms

Jan 2024

github.com/negarhonorvar/Bug-Algorithms

- Implemented Bug1, Bug2, and wall-following algorithms for GCTronic's e-puck in Webots to navigate a maze.
- Generated a maze map using Bug2 and split-and-merge algorithms.

Tron Game Agent

May 2023

github.com/negarhonorvar/TronGame

- Combined a genetic algorithm with Minimax as its fitness function for a real-time two-agent game.
- Built on a Unity framework based on Chillin's monitor games, with collision avoidance and competitive wall creation.

HONORS AND AWARDS

- **2025:** Among 10 outstanding computer engineering undergraduates at Shahid Beheshti University.
- **2020:** Ranked 2nd among 90 students admitted to SBU's Computer Engineering department.
- **2020:** Ranked 321st among 160,000 candidates in the national university entrance examination.
- **2019 & 2020:** First place in the first and second National Cyber Security and Cryptography Competitions, Ferdowsi University of Mashhad.
- **2017:** First place in the SAMPAD high-school entrance examination in Khorasan Razavi Province.

CERTIFICATES AND WORKSHOPS

2025	Unsupervised Learning, Recommenders, and Reinforcement Learning · DeepLearning.AI
2025	Advanced Learning Algorithms · DeepLearning.AI
2023	Supervised Machine Learning: Regression and Classification · DeepLearning.AI
2023	Building Web Applications in Django · University of Michigan
2023	Web Application Technologies and Django · University of Michigan
2023	Excel Basics for Data Analysis · IBM
2023	Introduction to Back-End Development · Meta
2018	Certificate of Excellence, Students' Vision on Vision Books Conference · SAMPAD
2018	15th National Mathematics Conference – Reza Sadeghi · SAMPAD

TEACHING ASSISTANT EXPERIENCE

Sep 2024– Feb 2025	Discrete Mathematics Lecturer: Dr. Farshad Safaei
Sep 2024– Feb 2025	Algorithms Design Lecturer: Dr. Ramak Ghavamizadeh
Feb–Jul 2024	Software Engineering Lecturer: Dr. Mehran Alidoostnia
Sep 2023– Jul 2024	Technical English Lecturer: Dr. Vahidi Asl
Sep 2023– Jan 2024	Computer Architecture · Dr. Rahmati Advanced Programming · Dr. Vahidi Asl Artificial Intelligence · Dr. Mehrnoush Shamsfard Compiler Design · Dr. Mehran Alidoostnia Computational Intelligence · Dr. Shahabedin Nabavi Operating Systems Laboratory · Dr. Shahabedin Navabi Statistics and Probability · Dr. Farshad Safaei